century," an assessment first articulated by the celebrated Italian physicist Danielle Amati.5 In a sense, then, it is as if our forebears in the late nineteenth century had been presented with a modern-day supercomputer, without the operating instructions. Through inventive trial and error, hints of the supercomputer's power would have become evident, but it would have taken vigorous and prolonged effort to gain true mastery. The hints of the computer's potential, like our glimpses of string theory's explanatory power, would have provided extremely strong motivation for obtaining complete facility. A similar motivation today energizes a generation of theoretical physicists to pursue a full and precise analytic understanding of string theory.

Witten's remark and those of other experts in the field indicate that it could be decades or even centuries before string theory is fully developed and understood. This may well be true. In fact, the mathematics of string theory is so complicated that, to date, no one even knows the exact equations of the theory. Instead, physicists know only approximations to these equations, and even the approximate equations are so complicated that they as yet have been only partially solved. Nevertheless, an inspiring set of breakthroughs in the latter half of the 1990s—breakthroughs that have answered theoretical questions of hitherto unimaginable difficulty—may well indicate that complete quantitative understanding of string theory is much closer than initially thought. Physicists worldwide are developing powerful new techniques to transcend the numerous approximate methods so far used, collectively piecing together disparate elements of the string theory puzzle at an exhilarating rate.

Surprisingly, these developments are providing new vantage points for reinterpreting some of the basic aspects of the theory that have been in place for some time. For instance, a natural question that may have occurred to you in looking at Figure 1.1 is, Why strings? Why not little frisbee disks? Or microscopic bloblike nuggets? Or a combination of all of these possibilities? As we shall see in Chapter 12, the most recent insights show that these other kinds of ingredients do have an important role in string theory, and have revealed that string theory is actually part of an even grander synthesis currently (and mysteriously) named M-theory. These latest developments will be the subject of the final chapters of this book.